Design and Control of a DC-DC Buck Converter with Soft-Starting Mechanism

Report submitted to the SASTRA Deemed to be University
as the requirement for the course

EEE320R01 : DIGITAL CONTROLLER FOR POWER ELECTRONIC APPLICATIONS

Submitted by

Shriram D
(Reg. No: 127159043)
Sridarshan B
(Reg. No: 127159009)

NOVEMBER 2025

SASTRA
ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act,1956)
THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR,TAMILNADU, INDIA-613401

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR-613 401

Bonafide Certificate

This is to certify that the report titled " **Design and Control of a DC-DC Buck Converter with Soft-Starting**" submitted as a requirement for the course, **EEE320R01: Digital Controller for Power Electronic Applications** for B. Tech. Electrical and Electronics Engineering (Smart Grid and Electric Vehicles) programme, is a bonafide record of the work done by **Mr.Shriram D(Reg. No. 127159043), Mr. Sridarsan B(Reg. No. 127159009)** during the academic year 2025-26, in the **School of Electrical and Electronics Engineering**, under my supervision.

Signature of Project Supervisor:

Name with Affliation : **Dr.Santhosh.T.K., (SAP / EEE / SEEE)**

Date : / 11 / 2025

Project *Vivavoce* held on

Examiner-I **Examiner-II**

SASTRA

ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION

DEEMED TO BE UNIVERSITY

(U/S 3 of the UGC Act, 1956)

THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

SCHOOL OF ELECTRICAL & ELECTRONICS ENGINEERING
THANJAVUR-613 401

Declaration

We declare that the report titled "**Design and Control of a Buck Converter with Soft-Starting mechanism** " submitted by us is an original work done by us under the guidance of **Dr. Santhosh.T.K.,SAP, School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the academic year 2025-26, in the **School of Electrical and Electronics Engineering**. The work is original and wherever We have used materials from other sources, We have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the candidate(s) :

Date **:** 14/ 11/ 2025

ACKNOWLEDGEMENTS

We express our gratitude to the Honourable Chancellor **Prof.R.Sethuraman** and our beloved Vice-Chancellor **Dr. S. Vaidhyasubramaniam**, SASTRA Deemed University, for the opportunity of pursuing our engineering studies in this esteemed institution and undertaking the project work.

We thank **Dr. S. Swaminathan**, Dean - Planning & Development, and **Dr. R. Chandramouli**, Registrar, SASTRA University, for granting permission and extending facilities to carry out this project.

We express our sincere thanks and gratitude to **Dr. K. Thenmozhi**, Dean, SEEE, and **Dr. A. Krishnamoorthy**, Associate Dean - Academics/SEEE SASTRA Deemed to be University, for their support in supporting the accomplishment of this work.

We also render our sincere thanks to the research scholars, **Mr. Saravanan**, **Mr. Saravanavel**, and **Mrs. Keerthiga** from the HFPC lab, for their kind technical support during this project. We also express our gratitude to the EMS lab at SASTRA - MHI Training Centre for providing support in fabricating our PCB boards.

We would like to thank our guide, **Dr. Santhosh T.K.**, SAP/SEEE, for his guidance and support, which ultimately led to the success of this project. His emphasis on making learning an experience allowed us to learn while making mistakes and rectifying them to learn, not only the scientific concepts behind power systems but also the process of analysing results.

We would like to thank our friends who have supported us. We would also like to thank the lab assistants, **Mr. Sembaiyya** and **Mr. Subash**, from SMTC-EMS Lab for their practical expertise and for providing the necessary software tools.

And finally, we would like to acknowledge the appreciation and support that our parents provided, ensuring we faced minimal obstacles throughout the project.

ABSTRACT

The demand for efficient and reliable DC–DC power conversion has increased with the rapid advancement of embedded and industrial electronic systems. This project focuses on the design and control of a buck converter that steps down a 12 V DC input to a regulated 3.3 V output. The converter integrates soft-start and dead-time protection features to ensure smooth startup, reduce inrush current, and enhance switching safety. The soft-start mechanism gradually ramps up the PWM duty cycle during power-up to prevent voltage overshoot, while dead-time control provides non-overlapping switching between the high-side and low-side MOSFETs, preventing short-through conditions. A Texas Instruments C2000 microcontroller is used to generate complementary PWM signals, which drive the MOSFETs through an IR2104 gate driver IC. The converter circuit was designed in KiCad and simulated in MATLAB/Simulink to verify output voltage and switching characteristics. Configuration and testing were performed in Code Composer Studio (CCS), validating efficient step-down conversion with smooth operation and stable performance suitable for embedded and low-voltage power applications.

Specific Contribution

- Designed and simulated the buck converter circuit in MATLAB/Simulink, verifying output voltage characteristics and analyzing converter performance.
- Configured system peripherals and tested the control code in Code Composer Studio (CCS) using the C2000 microcontroller, ensuring proper PWM generation, soft-start behavior, and dead-time functionality.

Specific Learning

- Gained knowledge in designing and simulating a buck converter using MATLAB/Simulink and configuring the C2000 microcontroller in CCS for testing soft-start and dead-time control functionalities.

Signature of the Guide **Student Reg. No:** 127159009

Name of the Guide: Dr.Santhosh.T.K(SAP/EEE/SEEE) **Name:** Sridarshan.B

ABSTRACT

The demand for efficient and reliable DC–DC power conversion has increased with the rapid growth of embedded and power electronic systems. This project focuses on the design and control of a digitally operated buck converter that steps down a 12 V DC input to a regulated 3.3 V output. The converter incorporates soft-start and dead-time protection features to ensure smooth startup, prevent inrush current, and enhance switching reliability. A Texas Instruments C2000 microcontroller generates complementary PWM signals to drive the high-side and low-side MOSFETs through an IR2104 gate driver IC, while an INA219 sensing module continuously monitors output voltage and current for closed-loop regulation. A PI controller implemented in the C2000 dynamically adjusts the PWM duty cycle to maintain a stable output under varying load conditions. The design and simulation were carried out using KiCad and MATLAB/Simulink, respectively, and the PCB layout was optimized for short current paths, minimal EMI, and improved thermal performance. The resulting converter achieved efficient voltage regulation, low output ripple, and reduced switching losses, making it suitable for embedded controllers, sensor systems, and low-voltage industrial applications requiring compact and reliable power supplies.

Specific Contribution

- Designed the complete buck converter schematic in KiCad and performed PCB layout routing, ensuring optimal component placement, signal integrity, and grounding.
- Carried out PCB soldering, assembly, and hardware testing, validating the converter's performance and functionality with soft-start and dead-time protection features.

Specific Learning

- Gained knowledge on how to design and develop a buck converter, including component selection, control strategy, and performance optimization in closed-loop configurations.
- Learnt the practical aspects of PCB design, routing, and hardware implementation, along with testing and analyzing converter performance using MATLAB/Simulink and embedded control systems.

Signature of the Guide **Student Reg. No:** 127159043

Name of the Guide: Dr.Santhosh.T.K(SAP/EEE/SEEE) **Name:** Shriram.D

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

Introduction

1.1 Overview

Efficient DC–DC conversion is essential in modern embedded and power electronic systems to provide stable and reliable power to low-voltage circuits. A buck converter (step-down converter) is widely used to obtain a regulated lower voltage from a higher DC source. A synchronous buck uses two Semiconductor switches to step down the input voltage by switcing the high side and low side MOSFETs alternatively.When high side MOSFET is ON,the inductor stores energy,when it is off the low side MOSFET provide a low resistance path for the inductor current.Here the output voltage can be controlled by varying the duty of the high-side MOSFET

The gain of the Buck Converter is given by Vout = Vin*Duty cycle

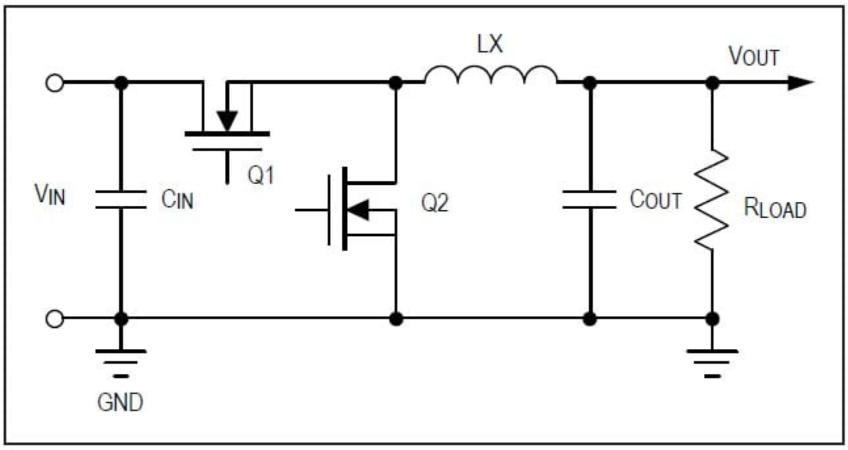

Figure 1.1: Cirucuit diagram of synchronous buck converter

This project focuses on the design and implementation of a closed-loop controlled buck converter that steps down 12V to 3.3V, suitable for powering microcontrollers and other low-voltage electronics. The design process includes simulation, PCB design, component selection, and hardware testing. closed-loop control improves voltage regulation, transient response, and efficiency compared to open-loop designs.

1.2 Need for a 12V to 3.3V Converter

Many embedded systems use 12V as the main supply while their logic and communication modules operate at 3.3V. Hence, a reliable and efficient 12V-to-3.3V converter is required to ensure proper functioning and energy efficiency.

Typical 3.3V-powered modules include:

- Microcontrollers (e.g., STM32, ESP32)
- Sensors and communication modules (Wi-Fi, Bluetooth, LoRa)
- Logic-level and control ICs in embedded devices

1.3 Applications

The 12V-to-3.3V buck converter finds use in several domains:

- **Industrial and Automation Systems:** Used in PLCs, control units, and sensor modules that require 3.3V logic from 12V supplies.
- **Electric Vehicles:** Powers low-voltage electronics such as BMS controllers, infotainment units, and sensors from the 12V auxiliary line.
- **Renewable Energy Systems:** Supplies power to control and monitoring circuits in solar inverters and IoT-based energy setups.

1.4 Project Summary

The project includes simulation of the buck converter, PCB layout design, hardware implementation, and closed-loop testing using the C2000 microcontroller. The converter successfully regulates a 3.3V output from a 12V input, maintaining stability under varying load conditions and validating the effectiveness of the feedback-based control approach.

CHAPTER 2

Objectives

2.1 Project Objectives

The primary objectives of this project are as follows:

- **Design and Implementation:** To design and build a closed-loop controlled buck converter capable of stepping down a 12V input to a regulated 3.3V output, suitable for powering low-voltage embedded systems and digital modules.
- **Voltage Regulation:** To achieve precise and stable output voltage across varying load conditions by employing feedback control mechanisms.
- **Efficiency Optimization:** To maximize the conversion efficiency and minimize output voltage ripple through careful converter topology selection, component choice, and PCB layout practices.
- **Simulation and Analysis:** To perform circuit simulation and theoretical analysis, validating system behavior before hardware fabrication.
- **PCB Design and Fabrication:** To develop a compact printed circuit board for the converter, select and source appropriate components, and assemble the hardware prototype.
- **Experimental Validation:** To test and verify the practical converter against technical criteria including output regulation, efficiency, transient response, and real-world applicability.
- **Application Demonstration:** To demonstrate the converter's utility for domains such as embedded systems, industrial automation, electric vehicles, and renewable energy setups, highlighting its relevance and adaptability.

These objectives are designed to ensure a comprehensive approach—covering theoretical, design, and practical aspects—to realize a robust and efficient regulated DC–DC converter for modern electronic systems.

CHAPTER 3

Experimental Setup and Analysis

3.1 Block Diagram and Explanation

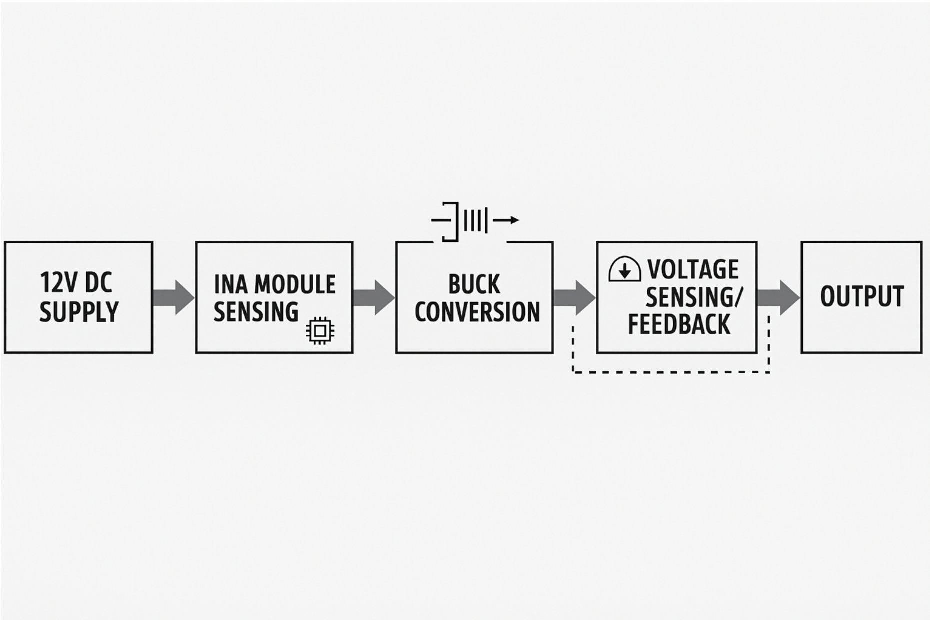

Figure 3.1: Block Diagram of Closed-loop buck converter using C2000

Explanation:

The buck converter system is designed to regulate output voltage using the C2000 microcontroller. This system is designed around efficient power management and monitoring. It begins with the input Supply, which provides the primary power source. An INA Module Sensing stage immediately follows, acting as a critical monitor to precisely measure the input current, voltage, and power consumption, providing essential data for system health and safety. The measured power then flows to the Buck Conversion stage, a crucial step that efficiently steps down the input voltage to the stable, lower level required by the final load. To ensure the output remains highly regulated, the system incorporates Voltage Sensing/Feedback. This stage continuously monitors the output voltage and sends an error signal back to the Buck Converter to dynamically adjust its operation, thereby maintaining a consistent and accurate voltage for the Output load, completing the robust and controlled power delivery cycle.

3.2 Simulated circuit and Design Equations

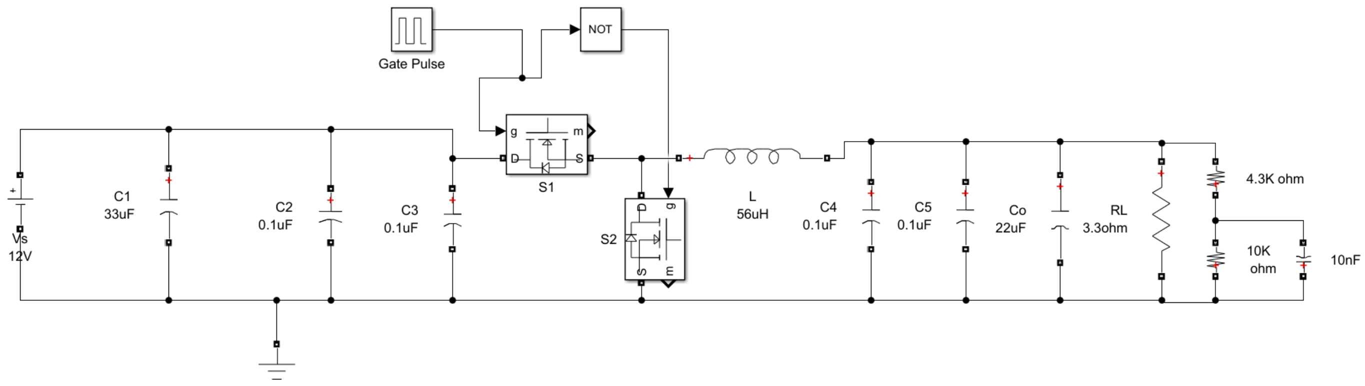

Figure 3.2: Circuit diagram of the closed-loop buck converter

Design Equations:

$$D = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{12}{3.3} = 0.275 \tag{3.1}$$

$$\Delta V_{\text{ripple}} = 0.01 \times V_{\text{out}} = 0.03 \tag{3.2}$$

$$I_{\text{out}} = \frac{V_{\text{out}}}{R_{\text{out}}} \tag{3.3}$$

$$R_{\text{out}} = \frac{3.3}{1} = 3.3\ \Omega \tag{3.4}$$

$$I_{\text{out}} = \frac{3.3}{3.3} = 1\ \text{A} \tag{3.5}$$

$$\Delta I_L = 0.25 \times I_{\text{out}} = 0.25\ \text{A} \tag{3.6}$$

$$V_o = D \times V_{in} \tag{3.7}$$

$$L = \frac{(V_{in} - V_o) \times D}{\Delta I_L \times f_s} \tag{3.8}$$

$$C = \frac{\Delta I_L}{8 \times f_s \times \Delta V_o} \tag{3.9}$$

This buck converter design steps down the input voltage from 12V to a stable 3.3V output, ideal for digital logic and embedded loads. Operating at a switching frequency of 150kHz, it employs a 33uH inductor calculated to maintain continuous conduction and keep the current ripple within practical limits. A 22uF output capacitor is selected for low output voltage ripple, while a 33uF input capacitor handles supply-side noise and stabilizes the input voltage. The design equations

shown establish the relationship between the duty cycle, inductor, and capacitor values based on voltage, current ripple, and frequency, guiding precise selection for reliable and efficient operation.
Passive components
Cin=33uF
Co=22uF
L=33uH

3.3 MATLAB Simulation and Results

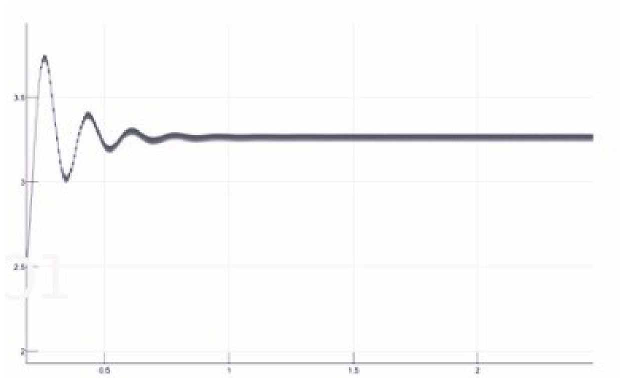

Figure 3.3: Output Voltage waveform

This MATLAB simulation of the buck converter demonstrates the effectiveness of the design and component choices. The top plot shows the output voltage settling quickly to 3.3V and maintaining a stable value, matching the target set by our calculations.

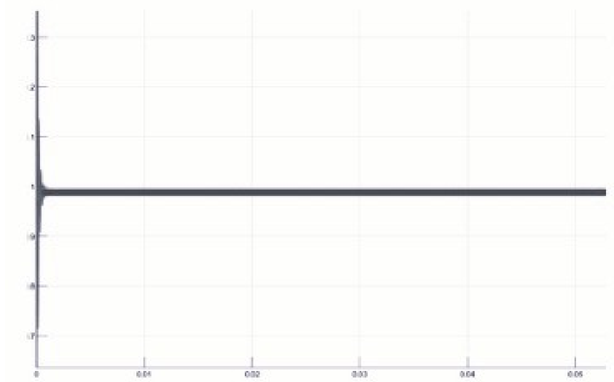

Figure 3.4: Inductor current waveform

The middle plot displays the output current holding steady right at the designed Iout of 1A , which confirms the load handling capability of the circuit

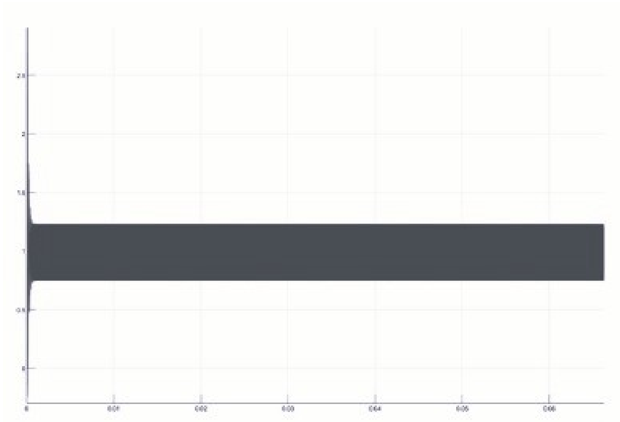

Figure 3.5: Inductor current waveform

•

The bottom plot illustrates the inductor current waveform: after a brief transient, it consistently operates with the expected ripple, staying well within the acceptable bounds calculated during component selection. The max saturation current of our selected inductor is 4A which is well above the peaking current here

Together, these results validate the design equations and specifications. The voltage ripple is minimized by the chosen output capacitor, and the inductor current shape matches theoretical predictions for 150kHz switching. The converter thus delivers reliable, regulated output performance suitable for modern embedded circuits as intended.

3.4 KiCad Schematic Design

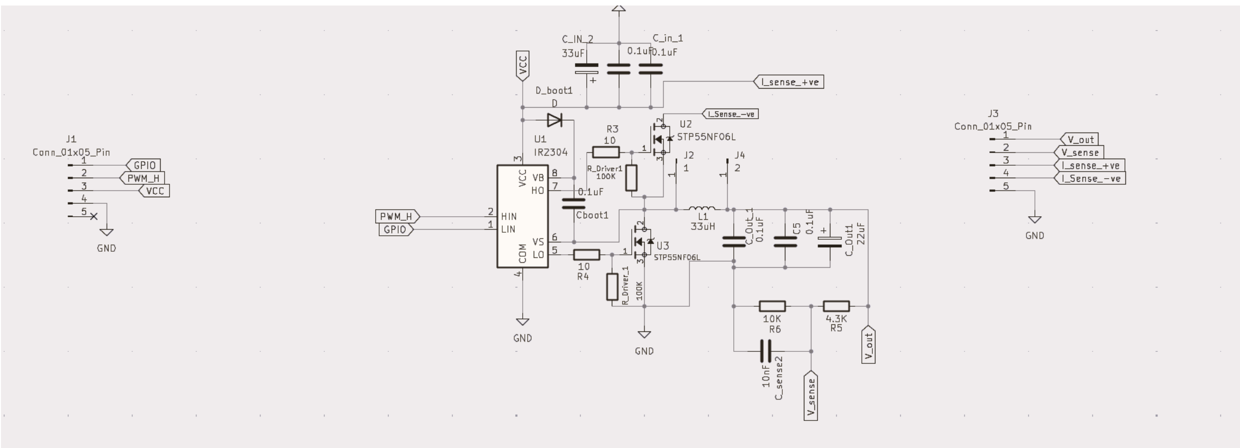

Figure 3.6: KiCad schematic of the buck converter

Table 3.1: Component Symbol and Footprint Assignments

Item	Symbol	Value	Footprint
1	C1 - C5	0.1μF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
2	Cboot1	0.1μF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
3	C_in_1 - C_in_2	0.1μF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
4	C_in_1 - C_in_2	33μF	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
5	C_Out1 - C_Out_2	22μF	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm
6	C_Out_1	0.1μF	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P2.50mm
7	C_sense2	10nF	Capacitor_THT:C_Disc_D4.3mm_W1.9mm_P5.00mm
8	D_boot1	D	Diode_THT:D-DO-41_SOD81_P10.16mm_Horizontal
9	J1 - J3	Conn_01x05_Pin	Connector_PinHeader_2.54mm:PinHeader_1x05_P2.54mm_Vertical
10	L1	L	Inductor_SMD:L_Coilcraft_MSS1278H-XXX
11	R3 - R4	10Ω	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
12	R5	$4.3\text{K}\Omega$	Resistor_THT:R_Axial_DIN0411_L9.9mm_D3.6mm_P12.70mm_Horizontal
13	R6	$10\text{K}\Omega$	Resistor_THT:R_Axial_DIN0411_L9.9mm_D3.6mm_P12.70mm_Horizontal
14	R_Driver1 - R_Driver_2	$100\text{K}\Omega$	Resistor_THT:R_Axial_DIN0414_L11.5mm_D4.5mm_P15.24mm_Horizontal
15	U1	IR2304	Package_DIP:DIP-8_W7.62mm
16	U2 - U3	STP55NF06L	Package_TO_SOT_THT:TO-220-3_Vertical

The KiCad schematic design details the implementation of the Buck Converter topology, integrating the driver IC U1 (IR2304) and the MOSFET switches U2/U3 (STP55NF06L) to efficiently step down the voltage. The circuit utilizes passive components—capacitors (Cboot1, C_in, C_out) and an inductor (L1)—to filter and smooth the power output. All components, from the power connectors (J1-J3) to the resistors (R3-R6), are assigned specific footprints (e.g., THT-axial, THT-disc) to ensure accurate and manufacturable PCB layout, as itemized in the accompanying Component Symbol and Footprint Assignments table.

3.5 Bill Of Materials

Table 3.2: Bill of Materials for the Closed-Loop Buck Converter

S.No	Component	MPN	Qty	Unit Cost ()	Total ()
1	Driver IC IR2304(U2)	IR2304PBF	2	213.57	427.14
2	Power MOS-FET(Q1,Q2)	STP55NF06L	4	156.31	625.24
3	Capacitor (0.1uF)	K104K15X7RF53L2	5	9.24	46.20
4	Capacitor (33uF)	63SXE33M	1	167.54	167.54
5	Capacitor (22uF)	A758EK226M1EAAE050	1	27.92	27.92
6	Resistor (10K)	MFR50SFTE52-10K	4	8.71	34.84
7	Resistor (10)	CFR-50JB-52-10R	4	9.58	38.32
8	Resistor (4.3K)	MF50 4K3	2	10.62	21.24
9	Freewheeling Diode	1N5819	2	7.03	14.06
10	Inductor (33uH)	MSS1278H-333MED	1	220.98	220.98
11	Connector (5 pin)	MC34655	2	10.05	20.10
12	Connector (3 pin)	MC34607	2	3.77	7.54
				Total	**1651.12**

3.6 PCB Layout and Component Placement

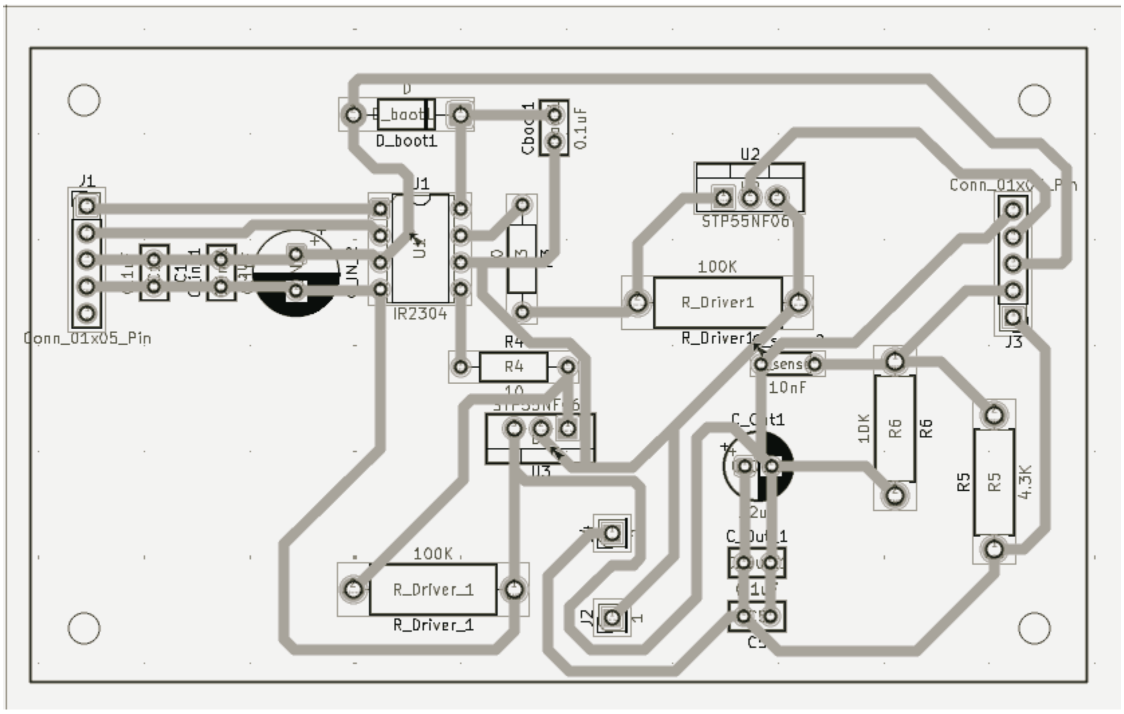

Figure 3.7: PCB layout of the closed-loop buck converter

Explanation:
The PCB layout was designed considering minimal loop area, short gate drive paths, and efficient heat dissipation. The driver IC IR2304 is placed at the center of the board to minimize connection lengths and ensure symmetrical connections between the high-side and low-side MOSFETs.Power routes are designed with 45 degree cuts and minimal branching to maintain even current flow.The 45 degree routing angles helps to prevent signal deflections,ensure smoother current transistions.The central ground area near IR2304 and MOSFET sources acts as the star grounding point.Power ground, driver ground, and signal ground all meet here independently.This ensures clean switching, accurate sensing, and low EMI, making the circuit stable and efficient.

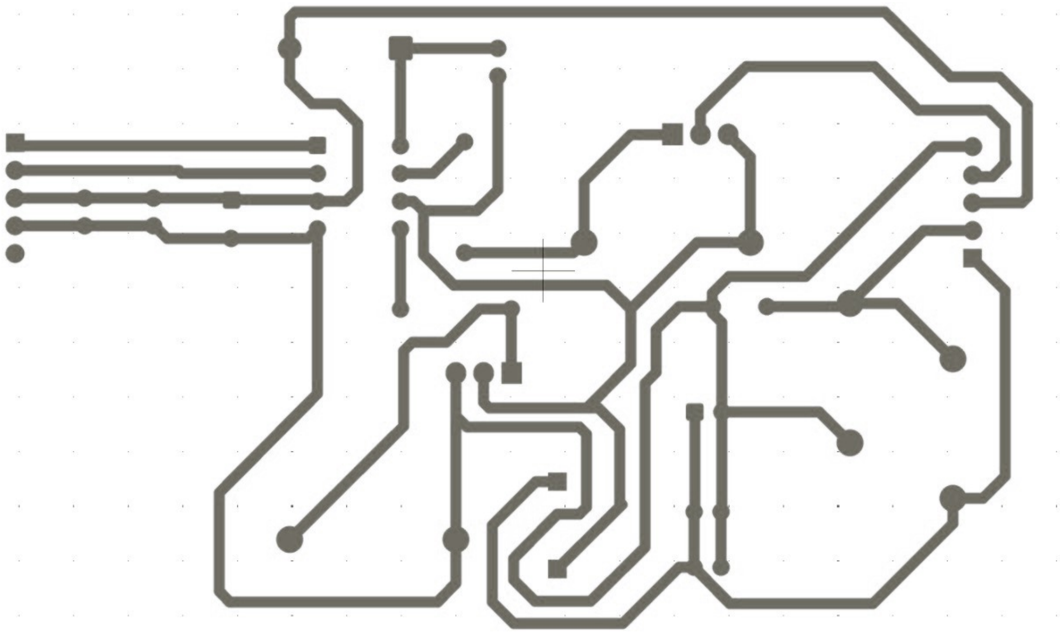

Figure 3.8: Back Copper View

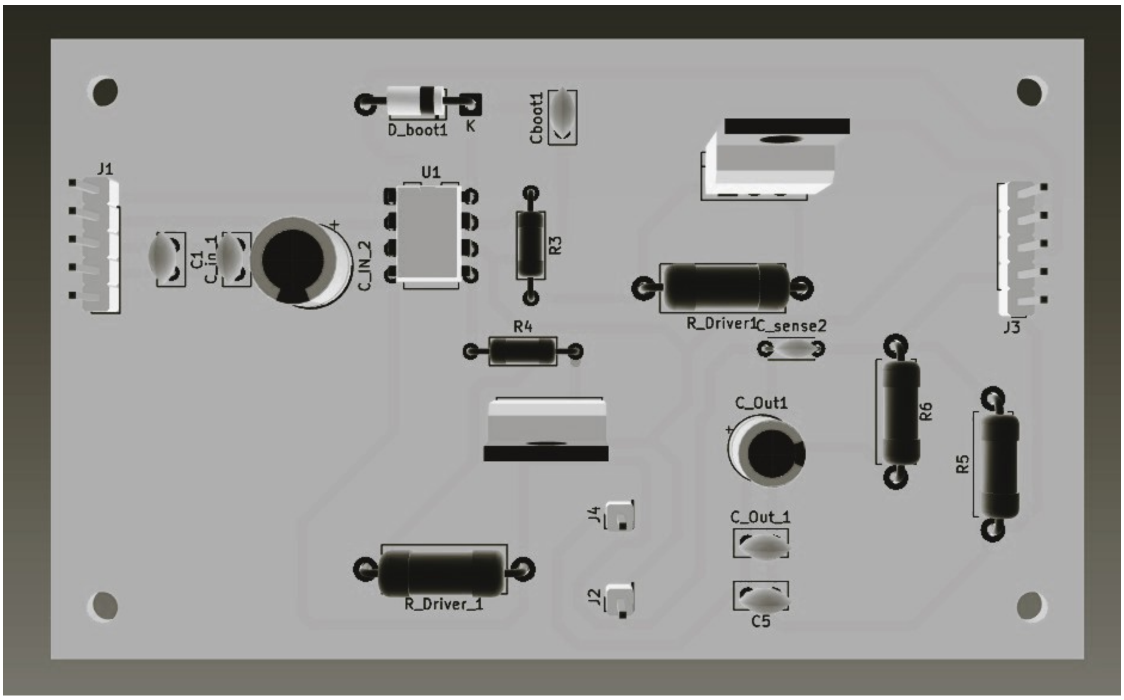

Figure 3.9: 3D-View

CHAPTER 4

Software Implementation Using C2000 Controller

4.1 SysConfig Initialization

EPWM

Time base Settings

We can set out required switching frequency by setting the Tbprd value.

$$T_{\text{BP}} = \frac{F_{\text{TBCLK}}}{2F_{\text{TBPWM}}}$$

$100^6/2 \times 150^3 = 333$ Time Period is set for a switching frequency of 150KHz. The appropriate calculated value of 333 is set. Counter mode - Upcount

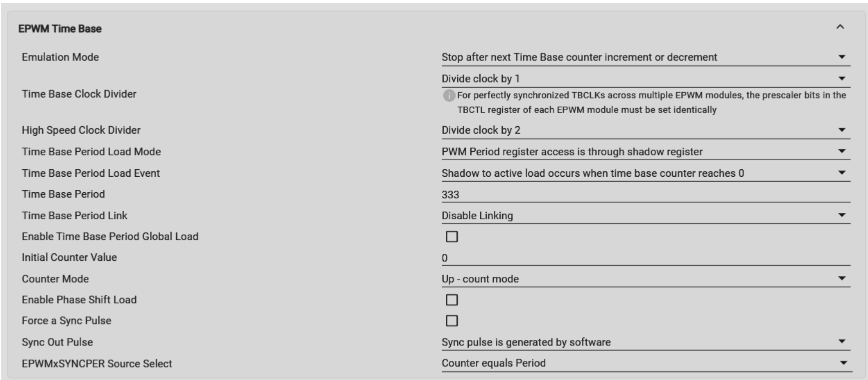

Figure 4.1: Time Base settings

Counter Compare

Counter Compare sets the Duty cycle of operation. According to our calculated duty cycle of 27 percent, we give it around 30 percent considering the losses. But, we do not set in the sysconfig itself, instead to apply the soft starting technique- to gradually rise the duty cycle we initialize the COMPA value in sysConfig as zero, a nd we program in the .c file to ramp it up Counter Compare values set to 0 to enable soft switching which slowly ramps up the duty cycle to the required value(120)

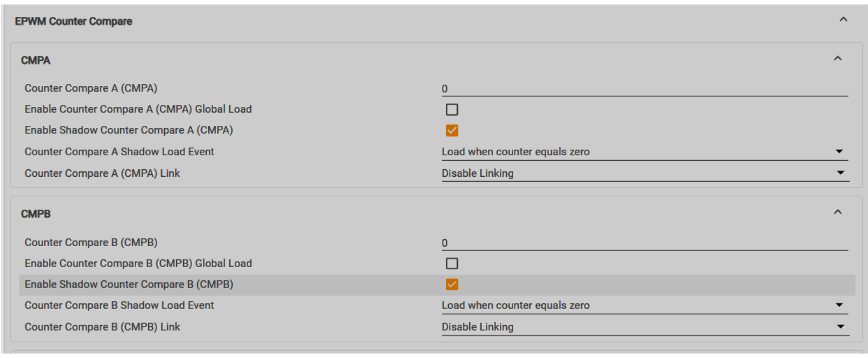

Figure 4.2: Counter Compare

Action Qualifier

Action Qualifier is the main decision making brain of the C2000, it decides the switching period and actions. We use EPWM2a for our signal. As we use the UP-Count mode, the PWM goes high at zero and becomes low once the timer matches with COMPA value.

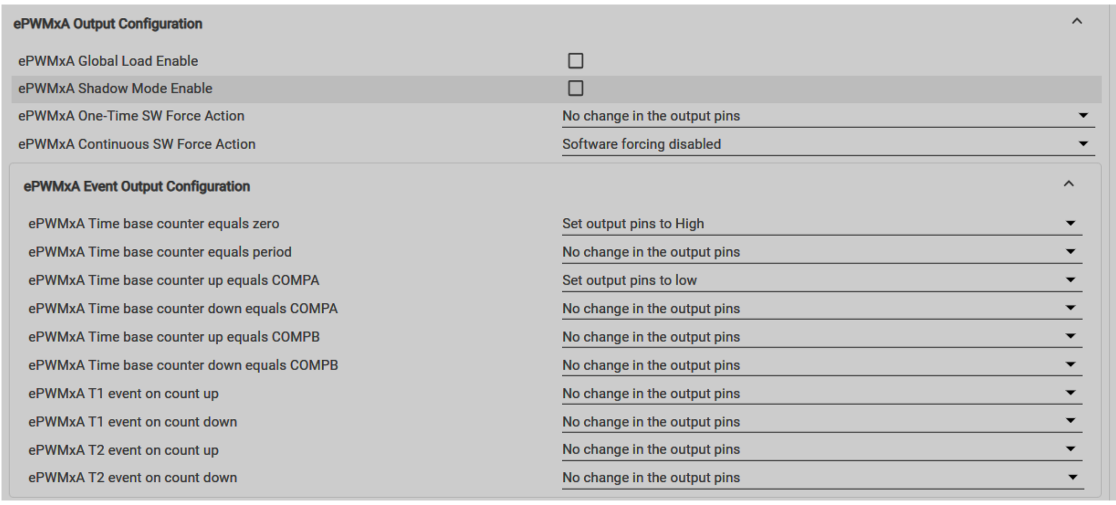

Figure 4.3: EPWMA

4.2 Programming in CCS

```
// Soft start / PI parameters
#define CMPA_TARGET          150U
#define SOFTSTART_STEP       2U
#define SOFT_TIMER_FREQ      1000U
#define DB_COUNTS            10U
```

These variables define how the soft-start should behave. `CMPA_TARGET` is the final PWM compare value we want to reach, and `SOFTSTART_STEP` decides how fast we ramp the duty cycle. `gCurrentCMPA` always holds the present PWM value, starting from zero, and `gSoftStartCompleted` simply indicates when the soft-start has finished.

```
// --------------------- EPWM_SafeInit ----------------------------
void EPWM_SafeInit(void)
{
    // Start with CMPA = 0 for safe startup
    EPWM_setCounterCompareValue(EPWM_BASE_EP, EPWM_COUNTER_COMPARE_A, 0U);
    gCurrentCMPA = 0U;

    // Dead-band configuration (override SysConfig counts if needed)
    EPWM_setRisingEdgeDelayCount(EPWM_BASE_EP, DB_COUNTS);
    EPWM_setFallingEdgeDelayCount(EPWM_BASE_EP, DB_COUNTS);

    // Ensure the EPWM action qualifiers and polarity are configured in SysConfig
    // TBPRD should match EPWM_TBPRD_COUNTS if you convert duty -> counts in ISR.
    EPWM_setTimeBasePeriod(EPWM_BASE_EP, EPWM_TBPRD_COUNTS);
}
```

Before the converter begins controlling anything, the PWM output is forced to zero duty cycle. This ensures the buck converter does not suddenly start with a large pulse that could stress the MOSFETs, inductor, or input supply. The soft-start ramp always begins from this known safe point.

```
// --------------------- cpuTimer0SoftStartISR ---------------------
__interrupt void cpuTimer0SoftStartISR(void)
{
    CPUTimer_clearOverflowFlag(CPUTIMER0_BASE);

    if(!gSoftStartCompleted)
    {
        uint32_t next = (uint32_t)gCurrentCMPA + SOFTSTART_STEP;
        if(next >= CMPA_TARGET)
        {
            next = CMPA_TARGET;
            gSoftStartCompleted = true;
        }

        gCurrentCMPA = (uint16_t)next;
        EPWM_setCounterCompareValue(EPWM_BASE_EP, EPWM_COUNTER_COMPARE_A, gCurrentCMPA);

        if(gSoftStartCompleted)
        {
            // stop soft-start timer once done
            CPUTimer_stopTimer(CPUTIMER0_BASE);
            CPUTimer_disableInterrupt(CPUTIMER0_BASE);
        }
    }

    // Acknowledge timer PIE/interrupt group (match with your project)
    Interrupt_clearACKGroup(SOFTSTART_TIMER_INT_ACK_GROUP);
}
```

Figure 4.4: Soft Starting

This interrupt runs at a fixed rate during startup. Every time it executes, it adds a small step to the PWM compare value. This creates a smooth ramp from zero duty cycle to the target value. Once the ramp reaches the final level, the function marks soft-start as complete and stops further updates. After this point, control is handed over to the regulator.

CHAPTER 5

Results

Input from C2000

We checked our PWM signal generated using C2000 which is sent to the driver IC IR2104 using a DSO which triggers the gate of the mosfets. The driver being a half bridge driver required only one PWM signal and it is equipped to provide the complementary pwm to the other mosfet by providing the required dead time. Our calculated 150KHz frequency and a duty cycle of 27.5 percent is being observed which proves its proper working.

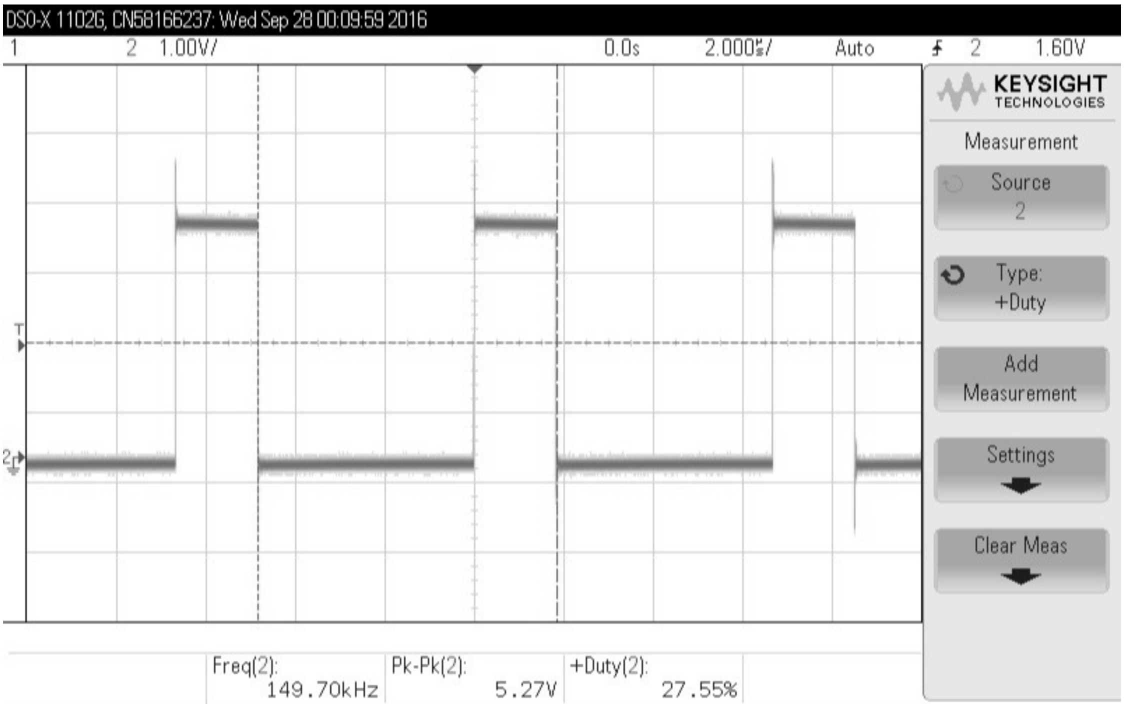

Figure 5.1: PWM input to the driver IC

5.1 PCB after soldering

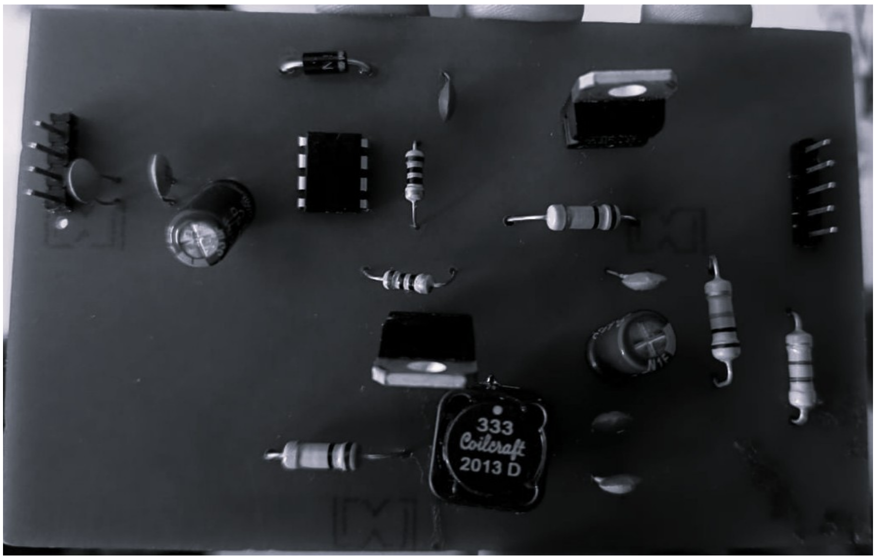

Figure 5.2: PCB Top View

After receiving the printed circuit board from fabrication, we performed manual soldering of the components using the lab facilities. After soldering, a continuity check was carried out to ensure that all components were placed and connected correctly, followed by functional testing. Tests performed: PWM input signal from C2000

Continuity test

Driver IC test

Load test

These tests were conducted before using the designed converter for the intended application. This ensures reliability and builds confidence in the performance of the system.

CHAPTER 6

Conclusion

The designed 12 V to 3.3 V buck converter was successfully implemented and tested. The hardware was fabricated on a custom PCB, followed by manual soldering and verification through continuity and functional checks. All preliminary tests—including the continuity test, driver IC test, and load test—were completed to ensure the reliability of the circuit before applying the final load conditions.

The converter achieved a stable output of 3.3 V at a load current of 1 A, matching the intended design specifications. The measured output ripple and current ripple remained within acceptable limits, confirming proper inductor sizing and capacitor selection. The switching waveform and inductor current were consistent with expected buck converter operation. The implemented soft-start prevented sudden inrush and ensured a smooth voltage rise without overshoot.

Overall, the converter performed reliably and met the required electrical characteristics. The results validate the design calculations, component selection, and control strategy used in this work. The successful hardware operation confirms that the system is suitable for the intended low-voltage DC applications and can be further extended or optimized for higher loads or improved efficiency in future work.

REFERENCES

1. Chandorkar, M. C. (2001). Digital control of switching power converters. *IEEE Transactions on Industrial Electronics 48*(2), 310–319.

2. Erickson, R. W. and D. Maksimovic (2007). Basic concepts of dc-dc converters. *IEEE Transactions on Power Electronics 22*(1), 2–16.

3. Instruments, T. (2015). C2000 microcontrollers for digital power conversion. *Application Report SPRABM6A 1*, 1–13.

4. Kazimierczuk, M. K. (2001). Pulse-width modulated dc-dc converters. *Wiley Encyclopedia of Electrical and Electronics Engineering 1*, 85–99.

5. Liu, L., S. Tian, D. Xue, T. Zhang, and Y. Chen (2019). Industrial feedforward control technology: a review. *Journal of Intelligent Manufacturing 30*, 2819–2833.

6. Lokesh, S. and N. R. Babu (2011). Design and implementation of digital control for buck converters using fpga. *International Journal of Electronics and Communication Engineering 4*(1), 1–7.

7. Mohammad, A. and O. Lutfi (2016). Soft-starting technique for dc-dc converters. *International Journal of Power Electronics and Drive Systems 7*(2), 242–252.

8. Rajesh, R., N. Prabaharan, T. Santhosh, R. Vadivel, and N. Gunasekaran (2024). A closed-loop using sampled-data controller for a new non-isolated high-gain dc-dc converter. *IEEE Transactions on Power Electronics*.

9. Ramkumar, R. and S. Anand (2018). Pcb design optimization for power electronics circuits. *International Journal of Engineering Research & Technology (IJERT) 7*(4), 350–356.

SIMILARITY CHECK REPORT

Team11

by D Uma

General metrics

15,667	2,280	320	9 min 7 sec	17 min 32 sec
characters	words	sentences	reading time	speaking time

Score

77

This text scores better than 77% of all texts checked by Grammarly

Writing Issues

84	68	16
Issues left	Critical	Advanced

Plagiarism

2 %

9 sources

2% of your text matches 9 sources on the web or in archives of academic publications